Tail Attacks on Web Applications (English translation)

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Tail Attacks on Web Applications

Huasong Shan† , Qingyang Wang† , Calton Pu † Computer Science and Engineering Division, Louisiana State University, Baton Rouge, LA, USA

{hshan1, qwang26}@lsu.edu College of Computing, Georgia Institute of Technology, Atlanta, GA, USA

calton@cc.gatech.edu

ABSTRACT 1 INTRODUCTION As the extension of Distributed Denial-of-Service (DDoS) attacks Distributed Denial-of-Service (DDoS) attacks for web applications to application layer in recent years, researchers pay much interest such as e-commerce are increasing in size, scale and frequency [1, 5]. in these new variants due to a low-volume and intermittent pattern Akamai's "quarterly security reports Q4 2016" [1] shows the spot- with a higher level of stealthiness, invaliding the state-of-the-art light on Thanksgiving Attacks, the week of Thanksgiving (involving DDoS detection/defense mechanisms. We describe a new type of the three biggest online shopping holidays of the year: Thanksgiv- low-volume application layer DDoS attack–Tail Attacks on Web ing, Black Friday, and Cyber Monday) is one of the busiest times of Applications. Such attack exploits a newly identified system vulner- the year for the retailers in terms of sales and attack traffic. Web ability of n-tier web applications (millibottlenecks with sub-second applications remain the most vulnerable entrance for any enterprise duration and resource contention with strong dependencies among and organization, so the attackers can exploit them to launch both

distributed nodes) with the goal of causing the long-tail latency low-volume and stealthy application-layer DDoS attacks. On the problem of the target web application (e.g., 95th percentile response other hand, in web applications especially e-commerce websites, time > 1 second) and damaging the long-term business of the ser- fast response time is critical for service providers' business. For vice provider, while all the system resources are far from saturation, example, Amazon reported that an every 100ms increase in the page making it difficult to trace the cause of performance degradation. load is correlated to a decrease in sales by 1% [24]; Google requires We present a modified queueing network model to analyze the 99 percentage of its queries to finish within 500ms [10]. Emerg- impact of our attacks in n-tier architecture systems, and numer- ing augmented-reality devices (e.g., Google Glass) need the associ- ically solve the optimal attack parameters. We adopt a feedback ated web applications with even greater responsiveness in order control-theoretic (e.g., Kalman filter) framework that allows attack- to guarantee smooth and natural interactivity. In practice, the tail ers to fit the dynamics of background requests or system state by latency, rather than the average latency, is of particular concern for dynamically adjusting attack parameters. To evaluate the practi- response-time sensitive web-facing applications [6, 10, 11, 18, 40]. cality of such attacks, we conduct extensive validation through In this paper we present a new low-volume application-layer not only analytical, numerical, and simulation results but also real DDoS attack–Tail Attacks, significantly worsening the tail latency cloud production setting experiments via a representative bench- on web applications. Web applications typically adopt n-tier archi- mark website equipped with state-of-the-art DDoS defense tools. tecture in which presentation (e.g., Apache), application processing We further proposed a solution to detect and defense the proposed (e.g., Tomcat), and data management (e.g., MySQL) are physically attacks, involving three stages: fine-grained monitoring, identifying separated among distributed nodes. Previous research on perfor- bursts, and blocking bots. mance bottlenecks in n-tier systems [41–43] shows that very short bottlenecks (VSBs) or millibottlenecks (with sub-second duration) CCS CONCEPTS with dependencies among distributed nodes not only cause queuing • Security and privacy Distributed systems security; Web delay in local tier, but also cause significant queuing delay in up- application security; Denial-of-service attacks; stream tiers in the invocation chain, which will eventually cause the long-tail latency problem of the target system (e.g., 95th percentile response time > 1 second). More importantly, this phenomenon usu- KEYWORDS ally starts to appear under moderate average resource utilization Long-tail latency; milli-bottleneck; n-tier systems; pulsating attack; (e.g., 50%) of all participating nodes, making it difficult to trace the web attack; DDoS attack cause of performance degradation. In the scenario of Tail Attacks, an attacker sends intermittent bursts of legitimate HTTP requests Permission to make digital or hard copies of all or part of this work for personal or to the target web system, with the purpose of triggering millibot- classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation tlenecks and cross-tier queue overflow, creating "Unsaturated DoS" on the first page. Copyrights for components of this work owned by others than ACM and the long-tail latency problem, where denial of service can be must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, successful for short periods of time (usually tens or hundreds of to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org. milliseconds), which will eventually damage the target website's CCS '17, October 30-November 3, 2017, Dallas, TX, USA reputation and business in the long term. © 2017

Association for Computing Machinery. The study of Tail Attacks complements previous research on ACM ISBN 978-1-4503-4946-8/17/10. . . $15.00 https://doi.org/10.1145/3133956.3133968 low-rate network-layer DDoS attacks [12, 14, 21, 22, 25, 27, 39],

Figure 1: Attack scenario and system model

low-volume application-layer DoS Attacks [8, 28], and flash crowds background requests and system state by dynamically tuning (usually tens of seconds or minutes) [19, 38, 44] which refer to the the optimal attack parameters. situation when thousands of legitimate users suddenly start to visit • Validating the practicality of our attacks through not only a website during tens of seconds or minutes due to a flash event (e.g., analytical, numerical, and simulation results but also real during the week of Thanksgiving). The uniqueness of Tail Attacks experimental results of a representative benchmark website from previous research is that Tail Attacks aim to create very short equipped with state-of-the-art DDoS defense tools in real (hundreds of milliseconds) resource contention (e.g., CPU or disk cloud production settings. I/O) with dependencies among distributed nodes, while giving an • Presenting a conceptual solution to detect and defense the "Unsaturated illusion" for the state-of-the-art IDS/IPS tools leading proposed attacks, involving three stages: fine-grained moni- to a higher level of stealthiness. toring, identifying bursts, and blocking bots. The most challenging task for launching an effective Tail Attack We outline the rest of this paper as follows. Section 2 describes is to understand the triggering conditions of millibottlenecks inside an attack scenario and the practical impact of Tail Attacks in real the target web system, and quantify their long-term damages on cloud production settings. Section 3 models our attack scenarios the overall system performance. To thoroughly comprehend the in an n-tier system using queueing network theory, and provides attack scenario, we exploit the traditional queueing network theory an effective approach to solve the potential optimal attack parame- to model the n-tier system, and analyze the impact of our attacks ters numerically. Further, we evaluate the attack analytical model to the end-users and the systems through two new proposed met- in JMT [7] simulator environment and suggest several guidelines rics: damage length during which the new coming requests can be to choose the optimal attack parameters in more complex cases dropped with highly probability, and millibottleneck length during (e.g., the competition for free slots of a queue between attack re- which the bottleneck resources sustain saturation. To fit the dynam- quests and normal requests, overloaded attack requests can be also ics of background requests and system state, we develop a feedback dropped by the front-tier server). Section 4 describes our concrete control framework. Given the guide of the proposed model and implementation to launch Tail Attacks in real web applications. We the implementation based on the feedback control algorithm, we adopt Kalman filter, a feedback control-theoretic tool, to automati- can effectively control the attacks, and find that our attacks can cally adjust the optimal attack parameters fitting the dynamics of not only achieve high attack efficiency, but also escape the detec- target system state(e.g., dataset size change) and background work- tion mechanisms based on human-behavior models, which further load. Section 5 shows our attack results of RUBBoS [34] benchmark increases the stealthiness of the attack. website we have conducted in real cloud production settings, which In brief, this work makes the following contributions: further confirm the effectiveness and stealthiness of the proposed attacks, and the practicality of the control framework in more prac- • Proposing Tail Attacks by exploiting resource contention tical

Web environments. Section 6 provides a “tit-for-tat" strategy with dependencies among distributed nodes, that can signif- to detect and defend our attacks targeting the unique scenario and icantly cause the long-tail latency problem in web applica- feature of the proposed attack. Section 7 discusses some additional tions while the servers are far from saturation. factors that may impact the effectiveness of Tail attacks. Section 8 • Modeling the impact of our attacks on n-tier systems based- presents the related work and Section 9 concludes the paper. on queueing network theory, which can effectively guide our attacks in an even stealthy way. 2 SCENARIO AND MOTIVATIONS • Adopting a feedback control-theoretic (e.g., Kalman filter) Attack Scenario. Consider a scenario of a Tail attack in an n-tier framework that allows our attacks to fit the dynamics of system in Figure 1. The detailed model analysis and experimental

95ile RT 98ile RT 99ile RT Avg. RT Param. Description

W/o Att. W/o Att. W/o Att. W/o Qi the queue size for the ith tier Setting Att. C i, A the capacity serving attack requests for the ith tier att. att. att. att. C i, L the capacity serving legitimate requests for the ith tier EC2-111-2K 255 1347 267 1538 308 1732 192 273 λi the legitimate request rate terminating in the ith tier EC2-1212-4K 247 1139 282 1341 328 1683 170 218 B the attack request rate during a burst L the burst length during a burst EC2-1414-6K 263 1085 270 1285 312 1628 160 206 V the burst volume during a burst Azure-111-1K 251 1153 274 1295 295 1821 176 290 T the interval between every two consecutive bursts Azure-1212-2K 1090 1284 1507 li the time to fill up the queue of the ith tier 254 278 295 177 221 PD the period of the requests dropped during a burst Azure-1414-3K 264 1090 297 1314 325 1510 181 242 PMB the period of a millibottleneck during a burst NSF-111-1K 141 1217 151 2262 159 7473 90 327 ρ(L) the average drop ratio during a burst ρ(T) the drop ratio during the course of an attack NSF-1212-2K 128 1101 143 1502 167 7016 88 309 NSF-1414-3K 118 1014 122 1413 127 3152 71 268 Table 2: Model parameters W/o att.: Without attacks, Att.: under Tail Attacks, RT: response time(ms)

Table 1: The measured long-tail latency under Tail Attacks in real cloud production setting. cloud platforms (Amazon EC2 [3], Microsoft Azure [29]) and one academic cloud platform (NSF Cloudlab[31]). We use a notation CloudPlatform-ServerTiers-BaselineWorkload to denote the cloud platform, the configuration of the n-tier system, and the background data of Tail Attacks are in the following sections. By alternating workload. For Server Tiers, We use a four-digit (or three-digit) nota- short “ON" and long “OFF" attack burst, an attacker guarantees tion #W#A#L#D to denote the number of web servers, application the attack both harmful and stealthy. Short “ON" attack burst is servers, load-balance servers (may not be configured), and database typically on the order of milliseconds. The following sequence of servers. More experimental details are available in Section 5.1. Ta- causal events will lead to the long-tail problem at moderate average ble 1 compares the tail latency of the target system under attack and utilization levels during the course of Tail Attacks. (Event1) The without attack, indicating the significant long tail latency problem attackers send intermittent bursts of attack but legitimate HTTP re- under attack. Such long tail latency problem (e.g., 95th percentile quests to the target system during the “ON" burst period; each burst response time > 1 second) is considered as severe performance of attack requests are sent out within a very short time period (e.g., degradation by most modern e-commerce web applications (e.g., 50ms). (Event2) Resource millibottlenecks occur in some node, for Amazon) [6, 10, 11, 18, 24]. At the same time, the average response example, CPU or I/O saturates for a

fraction of a second due to the time is still in acceptable range under attack, making the illusion burst of attack requests. (Event3) A millibottleneck stops the satu- of “business as usual" for system administrators. rated tier processing for a short time (order of milliseconds), leading to fill up the message queues and thread pools of the bottleneck tier, and quickly propagating the queue overflow to all the upstream 3 TAIL ATTACKS MODELING tiers of the n-tier system as shown in Figure 1b. (Event4) The fur- In this section, we provide a simple model to analyze the impact of ther incoming packets of new requests are dropped by the front tier our attacks to the end-users and the victim n-tier system. Based-on server once all the threads are busy and TCP buffer overflows in the the simplified model we introduce an effective approach of getting front tier. (Event5) On the end-user side, the dropped packets are the potential optimized attack parameters to achieve our attack retransmitted several seconds later due to TCP congestion control goal. Finally, we evaluate the model via simulation experiments (minimum TCP retransmission time-out is 1 second [16]), the end and suggest several approaches to tune the attack parameters. users with the requests encountering TCP retransmissions perceive very long response time (order of seconds). 3.1 Model Long “OFF" attack burst is typically on the order of seconds, in Queueing network models are commonly used to analyze the per- which the target system can cool down, clearing up the queued formance problems in complex computer systems [23], especially requests and returning back to a low occupied state shown in Fig- for performance sizing and capacity provision. Here, we use a well- ure 1a. Unlike the traditional flooding DDoS attacks which aim to tuned queuing network to model n-tier systems, and analyze the bring down the system, our attack aims to degrade the quality of sequence of causal events and the impact in the context of Tail service by causing the long-tail latency problem for some legiti- Attacks shown in Figure 1. Table 2 summarizes the notation and mate users while keeping the attack highly stealthy. The alternating description of the parameters used in our model. The basic attack short “ON" and long “OFF" attack burst can effectively balance the pattern (see Event1 in Section 2) shown in Figure 1 is that during trade-off between attack damage and elusiveness. the “ON" burst period (L) the attackers send a burst of attack re- Measured Long-Tail Latency. Table 1 shows the impact of Tail quests with the rate (B), after the “OFF" burst period (T-L) they send Attacks through concrete benchmark web application with real another burst again, and repeat this process during the course of production settings deployed in the most popular two commercial a Tail attack. If all the attack requests will not be dropped by the

target system (more complex case will be discussed in Section 3.3), of propagating the queue overflown. Finally, the required time to we can calculate the attack volume during a burst by: overflow all the queues in the n-tier system is the sum of l_i . Here, Q, C, and λ are constants. l_i is a function of B. $V = B \cdot L$ (1) Once all the queues are overflown (Event3 in Section 2) in the We assume that the external burst of legitimate HTTP requests n-tier system, the new incoming requests may be dropped by the (Event1 in Section 2) can cause sudden jump of resource demand front tier (Event4 in Section 2). We term the period of the requests flowing into the target system and cause millibottlenecks (Event2 dropped during a burst as damage length. If the attackers continue in Section 2) in the weakest point of the system [12]. In our model to send attack requests to the system with overflown queues, and analysis, we assume that the n-th tier is the bottleneck tier. For we assume that attack requests can always occupy the free position example, the bottleneck typically occurs in the database tier (the of the queue in the system (more complicated case will be

discussed n-th tier) in web applications due to the high resource consumption in Section 3.3), then we can approximately infer damage length by: of database operations. n X Due to the inter-tier dependency (call/response RPC style com- PD = L − li (5) i=1 munication) in the n-tier system, one queued request in a down- stream server holds a thread in every upstream server. Thus, the Further, the end-users with the dropped requests perceive very long system administrator typically configures the queue size of up- response time (Event5 in Section 2), leading to the long-tail latency stream tiers bigger than the queue size of downstream tiers. In this problem caused by our attacks (Event1 in Section 2), which can be case, millibottlenecks (Event2 in Section 2) caused by overloaded at- approximately estimated as follows, tack bursts can lead to cross tier queue overflow from downstream PD tiers to upstream tiers (Event3 in Section 2) due to the strong de- ρ(T) = (6) T pendency among n-tier nodes. If the queue size satisfies During a burst, the servers need to provide all the required com- (C1) Q 1 > Q 2 > ... > Qn−1 > Qn puting resources (including bottleneck resources) to serve both and the burst rate satisfies attack requests and normal requests. We term the period of a milli- (C2) λn + B > C n bottleneck during a burst as millibottleneck length, during which for all i=1,...,n, then the time needed to fill up the queue for the n-th bottleneck resources sustain saturation. Thus, millibottleneck length server is approximately should involve the resource consumption for both attack and nor- Qn mal requests during a burst. Equation (7) represents millibottleneck ln = (2) length derived through the geometric progression in mathematics (λn + B − C n,A) (more detailed derivation of this equation in Appendix A). (Q n−1 − Q n) l n−1 = (3) 1 1 (λn−1 + λn + B − C n,A) PMB = V C n, A (1 − (λn 1 (7) C n, L)) ... (Q 1 − Q 2) l 1 = Pn (4) where 1/Cn,A and 1/Cn, L are the service time for attack and normal (i=1 λ i + B − C n,A) requests in the bottleneck tier, respectively. When millibottlenecks occur in the n-th server, firstly the queue in the n-th tier is overflown during l n , which equals the available 3.2 Numerically Solve Attack Parameters queue size of the n-th tier divided by the newly-occupied rate for Based on the model, we can infer the damage and elusiveness of our the queue of the n-th tier in Equation 2. The available queue size attacks through damage length and millibottleneck length. Further, equals the queue size of each tier subtracting the queue size of if we assign the attack goal and know system parameters, we can its directly downstream tier. The newly-occupied rate equals the calculate the optimal attack parameters mathematically. incoming rate of each tier subtracting the outgoing rate of the Constant Parameters Estimation. To get some reasonable con- total system (Cn,A), the incoming rate of the n-th tier includes the stant parameters(λ,C i,A ,C i, N ,Q) in the model, we estimate these requests going through the n-th tier and terminating in the n-th constants via profiling the service time of each type of request of tier (B for the attack requests and λn for the normal requests). We each component tier in the benchmark web-site RUBBoS 34, the capacity of each tier C i can be calculated requests go through every tier and terminate in the last bottleneck from the service time. We choose heavy requests (e.g., long service tier (detailed implementation in Section 4.2). Equation 3 represents time by consuming more system bottleneck resource, detailed ex- the time to fill up the queue in the n-1-th tier. Because one queued planation in Section 4.2) as attack requests. Table 3 lists a group of request in a downstream server holds a position in the queue of reasonable values of the constants for our model profiled in RUB- every upstream server, after the n-th tier is full, the available queue BoS. During the profiling, we choose 2000 legitimate users with size of the n-1-th tier should be (Qn−1 - Qn). All the requests arriving 7-second think time as our baseline experiment. All the transactions to a downstream tier need to go through every upstream tier,

thus supported by RUBBoS are terminated in MySQL, each transaction the incoming rate of the n-1-th tier includes the request rate of follows a static page-load terminated in Apache, and no transaction terminating in the n-1-th tier (λ_{n-1}) and the request rate of going terminates in Tomcat. Thus, the request rate of each tier λ_i is 280, through the n-1-th tier ($\lambda_n + B$). Similarly, we can calculate the 0, and 280, respectively. We set the queue size of each server Q_i time to fill up every queue in the n-tier system during the process satisfying the condition C1 in Equation (2).

0.5 0.5 0.5 0.5

Burst Length L [s]

Burst Length L [s]

Burst Length L [s] Burst Length L [s]

0.4 0.4 0.4 0.4

0.3 0.3 0.3 0.3

0.2 0.2 0.2 0.2 PMB <= 0.5 PMB <= 0.5 PD >= 0.1 PD >= 0.1, Q=30 PD >= 0.2 PMB <= 0.3 0.1 PD >= 0.1 0.1 PD >= 0.1, Q=80 0.1 P >= 0.1 0.1 PMB <= 0.5 PMB <= 0.5 PD >= 0.1, Q=130 PDD>= 0.05 PMB <= 0.7 L <= 0.5 L <= 0.5 L <= 0.5 L <= 0.5 0 0 0 0 0 100 200 300 400 500 600 0 100 200 300 400 500 600 0 100 200 300 400 500 600 0 100 200 300 400 500 600 Attack Rate B [#/s] Attack Rate B [#/s] Attack Rate B [#/s] Attack Rate B [#/s] (a) There exists an overlapped fea- (b) As Q increases, the feasible (c) Various target P D . No solu- (d) Various target P M B . No so- sible zone below the dash line zone narrows down until no so- tion when P D is > 0.2s (red line), lution when P M B is < 0.3s (red P M B and above the solid line P D . lution when Q is 130 (red line). since no overlap exists. line), since no overlap exists.

Figure 2: Numerically solve the optimal attack parameters. Solid line depicts damage length, the target zone is above the solid line; and dash line depicts millibottleneck length, the target zone is below the dash line.

Server Tier (i) λ_i Ci, A Ci, L Qi shown in Figure 2a. In real cloud production settings, the queue size must be diverse according to the capacity of the websites. Apache 1 280 3443 3657 55 In Figure 2b, we can see that as the queue of the front tier (e.g., Tomcat 2 0 1300 1987 25 Apache) increases from 30 to 80, the feasible region reduces; when MySQL 3 280 280 725 6 it increases to 130 (the red line), the two inequations do not overlap, Table 3: Constant parameters estimation profiled in RUB- which implies that there is no solution to satisfy our predefined BoS experiment with 2000 concurrent users. attack goal. The fundamental reason is that our attack goal is too strict, which seems to be an impossible mission. Note that when the queue of the front tier is 80, in the strictest attack target cases (P D

> = 0.2 seconds, the red line in Figure 2c; or P M B <= 0.3 seconds,

Attack Goal and Solver. Suppose that we set our attack goal as the red line in Figure 2d), there is also no solution that can solve 95th percentile response time longer than 1 second which is a severe the attack parameters of our attacks. We will further discuss how long-tail latency problem for most e-commerce websites [6, 10, 11, to deal with the non-solution cases in Section 4.1. 18], and the duration of a millibottleneck less than 0.5 seconds in the bottleneck tier such that the

average utilization can be at moderate 3.3 Simulation Experiments level (e.g., 50-60%) to bypass the defense mechanisms. If we assume the burst interval T is 2 seconds, then the input to the model can The numerical solver in the previous section does not consider many be two inequations: damage length P D is bigger than 0.1 seconds aspects of the real system (e.g., the competition of the free position and millibottleneck length P M B is less than 0.5 seconds. Further, in the queue between attack requests and normal requests, over- we turn these two inequations to L as a function of B, the others loaded attack requests can be dropped, etc.). To further validate the parameters are all constants(Inequation (8) and (9)). simple model, we present results from Java Model Tools(JMT) [7] in which such limitations are absent. JMT is an open source suite for n modeling Queuing Network computer systems. It is widely used in X Qn L >= l i + 0.1 = the research area of performance evaluation, capacity planning in i=1 (λn + B − C n,A) n-tier systems. Thus, it is a natural choice to evaluate the impact of (Q n−1 − Q n) our attacks in n-tier systems. We modify the JMT code and simulate

- (8)

(λn−1 + λn + B − C n,A) the bursts of attack requests for our attacks with the configurable (Q 1 − Q 2) attack parameters in our model. +... + Pn + 0.1 (i=1 λ i + B − C n,A) Given the proposed model and the idea of solving the nonlin- ear optimization problem, we can get the feasible region of attack 1 C n,A L <= 0.5 (1 − λn) (9) parameters. We initialize the parameters in JMT similar to the set- C n, L B ting of our numerical solver, and choose a potential optimal point Thus, the problem of selecting a set of optimal attack parameters (400,0.215) in Figure 2a as our attack parameters, the attack rate B (B,L,V,T) can become a nonlinear optimization problem. Although is 400 requests per second and the burst length L is 0.215 seconds, nonlinear optimization problem is hard to solve since there exist so the attack volume per burst V is 86 (see equation (1)) if all the multiple feasible regions and multiple locally optimal points in attack requests will not be dropped by the target system. those regions, we can add more constraints to narrow the range of Results in JMT Figure 3 shows the results of one burst during feasible regions. For instance, the burst length obviously should be 1 second time period using fine-grained monitoring (e.g., 50 mil- less than target millibottleneck length(e.g.,L <= 0.5). liseconds) in JMT expriment. Figure 3a illustrates the process of Substituting the constant parameters in Inequation (8) and (9), we filling up all the queues in the n-tier system. Note that the queue of can get an unique feasible region as the potential attack parameters MySQL, Tomcat, and Apache is overflow from down-stream tiers

Apache Tomcat MySQL Apache Tomcat MySQL 6

Dropped Req. [#] CPU Usage [%] Queue Usage [#]

Attacker 60 100 Normal User 80 3 30 60 40 20 0 0 0 0 0.2 0.4 0.6 0.8 1 0 0.2 0.4 0.6 0.8 1 0 0.2 0.4 0.6 0.8 1 Timeline [s] Timeline [s] Timeline [s] (a) Process of filling up the queues in (b) Millibottleneck length (MySQL CPU) (c) The shift of the amount of dropped re- 3-tier system, queue overflow are propa- is less than the expected value 500ms, quests during damage length shows the gated from the bottleneck tier (MySQL) to since overloaded attack requests are also competition for the available slot of the the upstream tiers (Tomcat, then Apache). dropped by the front tier (Apache). queue freed by the outgoing requests.

Figure 3: Results of one burst during 1 second period in JMT experiment

to up-stream tiers overtime. The CPU saturations of the bottleneck Attack Para. Legitimate Users tier MySQL last approximately 400 milliseconds as shown in Fig- B L Dropped Reqs. Total Reqs. ρ(T) ure 3b, less than the expected value 500 milliseconds calculated 300 0.285 1096 55998 0.0196 by our model, since overloaded attack requests are also dropped 400 0.215 1099 54901 0.0200 by the front tier which will not go through the front tier and into 500 0.173 1705 54292 0.0314 the bottleneck tier. Figure 3c shows dropped requests perceived 600 0.144 2082 53915 0.0386 by attackers and legitimate users in the corresponding burst. Note 700 0.123 2326 53671 0.0433 that the dropped requests span two sampling duration (100 millisec- 800 0.108 2399 53598 0.0448 onds), validating our model expectation for the dropped length. The Table 4: Fix burst volume V =86. Bigger B Higher ρ(T), imply- interesting observation is that the dropped requests from attackers ing higher competition ability for the burst with larger B. is bigger than ones from legitimate users at the time of 0.15, the opposite phenomenon happens in the next sampling windows at the time of 0.2. This implies that the requests from the attacker and ones from the legitimate users compete the available position of the other three parameters (B,L,V). Due to interdependent relation- the queue freed by the outgoing request in the n-tier system during ship of these three parameters in Equation 1, we fix one parameter damage period [28], eventually the loser will be dropped. (L or V), then observe the impact with various attack rate B. We still However, when we aggregate the data of the legitimate users consider the marked potential optimal point (400,0.215) in Figure 2a during the 3-minute simulation experiment, the amount of dropped as our baseline attack parameters. requests is 1099 and the total requests is 54901, thus the actual drop First, we fix burst volume V as 86, and select a set of attack ratio for the legitimate users is 2%, which is far from the predefined rate (from 300 to 800) to conduct our attack experiments using JMT. goal of the drop ratio 5%. From this observation, we should calibrate Table 4 shows that as the attack rate increases, the drop rate ρ(T) ac- the drop ratio from Equation 6 to cordingly increases, which confirms that higher attack request rate, P D ρ(L) ρ(T) = (10) compared to normal request rate, can achieve higher competition T ability to seize the available position in the queue. Here, ρ(L) refers to the average drop ratio for the requests of the Next, we fix burst length L as 0.215 seconds, and select another legitimate users during damage length, which represents the com- set of attack rate to conduct our JMT experiments. Figure 4 depicts petition ability for attack requests compared to normal requests. In ρ(T) and ρ(L) as a function of the ratio of attack rate and service the previous JMT experiment, ρ(L) is approximately 0.4. rate for the bottleneck tier. We mark two vertical lines to split up Tuning the Attack Parameters. Due to the existence of the com- into three zones with various attack rate B. (1) In zone a, B is less petition between the attackers and the legitimate users, the attack- than C n,A , the drop ratios are all zero, since the attack rate is too ers may fail to get the predefined attack goal (e.g., 95th percentile low to trigger an effective millibottleneck to lead to cross tier queue response time > 1 second) by using the recommended attack param- overflown in the target system, which violates the condition C2 eters of the proposed model. We further investigate how to increase in Equation 2. (2) In zone b, B is bigger than C n,A , the drop ratio the competition ability of attack requests, and the drop ratio of the increases non-linearly as B increases. Observe that ρ(L) of Normal requests from the legitimates users during damage length, namely Users is a little bit bigger than ρ(L) of Attackers, because B is bigger ρ(L). Finally, the attackers can choose the optimal attack parameters than λn . In this case, the requests from the attackers can seize to achieve high damage with low cost and high stealthiness. For the available position in the queue with a more highly probability

simplicity, we assign attack interval T as fixed value (say, 2 seconds), than ones from the legitimate users. (3) In zone c, B is bigger than since our focus on this paper is to investigate how to effectively C 1,A , the attack requests are directly dropped by the most front trigger the millibottlenecks which is predominantly determined by tier, thus the increase of B does not contribute to the drop ratio of

Normal Users Normal Users Attackers 0.12 a b c 1 a b c 0.1 0.8 0.08 0.6 0.06 ρ(T)

ρ(L) 0.04 0.4 0.02 0.2 0 0 0 5 10 15 20 0 5 10 15 20 B/Cn B/Cn

Figure 4: Fix burst length L=0.215 seconds. (1) in Zone a, B < Cn, B is too low to trigger effective millibottlenecks; (2) in Zone b, B > Cn , the bottleneck is in n-th tier, ρ(T) increases as B increases; (3) in Zone c, B>C1, the attack requests are directly dropped by the most front tier, no obvious attack effect (ρ(T) is flat) even though B increases tremendously, implying that we should choose a moderate B until the attack goal is achieved (e.g., the green horizontal line is a target).

Figure 5: A feedback control framework

the legitimate users, it only increases the drop ratio itself. Given time, either it can not trigger millibottlenecks (not enough attack this observation, we should choose a moderate B until the attack requests) or it might trigger the defense alarm of the target system goal is achieved (e.g., the green horizontal line in Figure 4). (too frequent or massive attack requests). How can we dynamically adjust the attack parameters catering to instant system state and baseline workload is a big challenge for Tail Attacks. The static 4 TAIL ATTACKS IMPLEMENTATION attack parameters in the dynamical environment may make the attack either invalid like a "mosquito bite" or easily exposed to the 4.1 Overview detection mechanisms. In this section, we implement a feedback As mentioned before, the analytical model used in the numerical control-theoretic (e.g., Kalman filter) framework that allows attack- solver analyzes the simple scenario skipping many aspects of the ers to fit the dynamics of background requests or system state by system reality (e.g., the competition for the free position of the dynamically adjusting the optimal attack parameters in Figure 5. queue between attack requests and normal requests, overloaded at- Via our best practice, we find that the attack rate should not tack requests can be dropped, etc.), and the simulation experiments be invariable to maximize the attack effectiveness and stealthi- show more complicated cases involving the absent system real- ness. [25] suggests a double-rate DoS steam to minimize the at- ity. However, our attacks do not consider more realistic case with tack cost. We design a three-stage transmitting strategy to send dynamics of baseline workload 1 or system state(e.g., dataset size one burst: Quickly-Start , Steadily-Hold and Covertly-Retreat. In change). For example, the peak workload occurs at approximately Quickly-Start stage, the attacker sends the burst of requests at a 1:00 p.m. during the week of Thanksgiving [1]. A set of effective high rate to quickly fill up all the queues in the n-tier system, heavy attack parameters of Tail Attacks may become failed ones over requests (detailed explanation in Section 4.2) are preferred because it can consume more bottleneck resources and occupy the queue 1 For e-commerce applications, the baseline workload during the day time is usually longer with low cost and high stealthiness, the amount of heavy significantly higher than that during the mid-night period.

Algorithm 1 Pseudo-code for the control algorithm Apache Tomcat MySQL 400 1: procedure AdaptAttackParameters 2: AttackReqST EstimateServiceTime Queue Usage [#]

300 3: DamLen EstimateDamageLenByProber 4: MBLen EstimateMilliBottleneckLenByBots 200 Start Hold(Damage Length) Retreat 5: if DamLen = 0 then 6: /* can not fill up queue, increase B */ *100 Apache queue size 7: B B + stepB 8: else 0 9: gapDamLen Abs(DamLen - targetDamLen) Millibottleneck Length 0 0.2 0.4 0.6 0.8 1 10: stepV gapDamLen/AttackReqST 11: if DamLen > targetDamLen then Timeline [s] 12: /* reduce damage length by decreasing V */ Figure 6: Queue shifts during a three-stage burst(quickly- 13: V V − stepV start, steadily-hold, covertly-retreat) 14: else if DamLen < targetDamLen then 15: /* increase damage length by increasing V */ 16: V V + stepV requests during this stage should be large enough to temporar- 17: else ily saturate the bottleneck resource in the target system [40]. In 18: /* current values are the optimal parameters. */ Steadily-Hold Stage, the attacker should guarantee the queue can 19: end if be overflown during this stage and attack requests can seize the 20: end if free position in the queue with highly probability. Heavy requests 21: if MBLen > targetMBLen then are not necessary to serve as attack requests during this stage. We prefer light requests to hold on queue, such that in the last Covertly- 22: /* set max V */ Retreat stage, the attack requests can quickly and covertly leave 23: Vmax targetMBLen/AttackReqST the systems. In Covertly-Retreat stage, there is no attack request to 24: V Vmax be sent out. Figure 6 demonstrates the queue shifts during a three- 25: /* choose less heavy requests as attack requests */ stage burst. Through the strategy of variable attack rate and various 26: end if attack requests, we can solve the insolvable cases in Section 3.2 by 27: end procedure carefully choosing less heavy requests as attack requests. Return back to our model in Equation 5, we can see the rela- tionship between P D and the attack rate B were nonlinear. We can equals the difference between end-time and start-time. Typically, transfer P D as a function of V and B using equation 1: the end-to-end response time of a HTTP request involves three n V X parts: the network latency between the client and the target web PD = − li (11) B i=1 application, the queuing delay in the n-tier system, and the service time of each server. The network latency can be measured using If we fix the attack rate B, mathematically, P D and the attack vol- the ping command. When the target system is at low utilization, the ume V have a linear relationship; the same as P M B (see Equation 7). queuing effect inside the target system can be ignored. Thus, we These linear relationships provide us with a firm theoretical founda- can approximately estimate the service time of any HTTP request tion to dynamically adapt the optimal attack parameters fitting the supported by the target web system as the end-to-end response changes of system state and baseline workload. The overall control time subtracting the network latency when the target system is in algorithm can be described in Algorithm 1. the time block with a low workload. Since the service time of the estimated request may drift over time (e.g., due to changes in the 4.2 Estimator data selectivity and the network latency variation) in real appli- The Estimator, as illustrated in Figure 5, estimates three critical cations, we measure the service time of a HTTP request multiple metrics in the control algorithm implementing the proposed model: times and take the average. service time of the requests, damage length P D , and millibottleneck Previous research results [41] show that the predominant part of length P M B . We use a prober to monitor attacks and infer P D , the service time of a request is spent on the bottleneck resource in coordinate and synchronize bots to launch attacks and infer P M B . the system. We call the requests that heavily consume the bottleneck Estimating Service Time. Service time of a HTTP request is the resource as heavy requests with long service time (e.g., the request time that the target web system needs to process the request without

querying multi-tables in the database) while those consume no or any queuing delay. It is easy to calculate the end-to-end response little bottleneck resource as light requests with short service time time of a request using two time stamp of sending requests and (e.g., static requests) [40]. Thus, the prober naturally exploits light receiving responses [26], we term them start-time and end-time requests to monitor the impact of the attacks since it can be more of a HTTP request. The end-to-end response time of a request elusive under the radar without causing any alert of the target web

service time. In this way, the result overcharges a length of the network latency. The other option of inferring P M B is end-time of the last non-dropped attack request subtracting end-time of the first non-dropped attack request during one attack interval shown in Figure 7b. In this way, it undercharges a length of the service time. For the concrete environments of a special website, the attacker can choose any approach to estimate millibottleneck length. In our eval- uation section, we choose the second one, since the network latency is much longer than the service time in our RUBBoS environment. (a) Estimate P D by start-time (b) Infer P M B by end-time of of the last dropped probing the last non-dropped attack re- 4.3 Controller request subtracting start-time quest subtracting end-time of So far, we discuss many aspects that can influence the effectiveness of the 1st dropped probing re- the 1st non-dropped attack re- of our attacks. In the model, the competition for the free slot of the quest during an attack burst. quest during a interval. queue between attack requests and normal requests may impact precision of damage length, and the dropped attack requests may Figure 7: Demonstration of inferring damage length and mil- decrease millibottleneck length. For Estimator, the network latency libottleneck length by Estimator. variation and the drifted target system state might reduce the accu- racy of inferring damage length and millibottleneck length in our implementation. All of aspects lead to the observing and measuring system; and the bots can take heavy requests as candidate attack inaccuracy, and result in the invalidation of launching an effective requests since it can be more efficient causing millibottlenecks and attacks using our control algorithm. To mitigate these negative im- cross tier queue overflow. More advance technology about profiling pacts for our control algorithm, we adapt a popular feedback-based heavy requests will be discussed in Section 7. control tool, Kalman filter [20]. On the one hand, it can take past Through profiling and exploiting heavy requests [40], Tail At- measurements into account for implementing our feedback control tacks can transiently saturating the critical bottleneck resource (e.g., algorithm, reducing the impact of the process noise (e.g., baseline database CPU) of the target systems, which can saturate the critical workload and system state). On the other hand, it can mitigate the resource of the system with much lower volume (thus less bots measurement noise due to inaccuracy of the estimator. are needed) compared to that of traditional flooding DDoS attacks Let z(k) be the measurement of PD in k-th burst by Estimator. which usually try to fully saturate the target network bandwidth. Since PD is a linear function of burst volume V (see equ. (11)), we Estimating damage length P D . To estimate P D , the prober needs can define x(k) using a linear dynamical system model: to send probing requests (e.g., light requests) to the target web system at a predefined rate and record start-time and end-time of SystemDynamics : $x(k) = x(k-1) + U(k) + v(k)$ (12) a probing request. The recommended sending interval of probing requests is less than the target damage length, in the case, any MeasurementDomain : $z(k) = x(k) + w(k)$ (13) probing request may not miss the period of overflown queue caused by an attack burst and the prober can

sense P D [22]. Figure 7a where the variables v(k) and w(k) are the process noise (e.g., dynam- illustrates the implementation approach to estimate P D as start- ics of baseline workload) and the measurement noise (e.g., imperfect time of the last dropped probing request subtracting start- time of estimation by Estimator), respectively. U(k) is the expected control the first dropped probing request during a burst. Since some probing result, a linear function of burst volume V. requests may probably seize the free position in the queue and are Let $\hat{x}(k \mid k-1)$ be a priori estimate of state parameter x at burst k- not be dropped during the damage length, we can calibrate P D by th given the history of all k-1 bursts, and let $\hat{x}(k \mid k)$ be a posteriori multiplying ρ(L) (see equation 10). Some websites may send some estimate of state parameter x at k-th burst. Further, let $P(k \mid k)$ be alarm to the users if they send the requests at a very high rate. In a posteriori error covariance matrix which quantifies the accuracy this case, the prober can send the probing requests at an acceptable of the estimate $\hat{x}(k \mid k)$. The Kalman filter executes recursively for rate for the target web system and estimate the drop ratio during a each new observation including two phases: Predict in which a sampling period, then our control algorithm can exploit drop ratio priori estimate of state and error matrix are calculated, and Correct as the target criterion to dynamically adjust the attack parameters. in which a posteriori estimate of state and error matrix are refined Estimating millibottleneck length P M B . After sending a burst using the current measurement. The Kalman filter model for our of attack requests (e.g., heavy requests) to the target web system, control framework is given by: the bots can record start-time and end-time of an attack request Predict (Time Update) and estimate P M B . We only count the non-dropped attack requests, $\hat{x}(k \mid k-1) = \hat{x}(k-1 \mid k-1) + U(k)$ (14) since the dropped request involves TCP retransmission time-out. There are two options to infer P M B . One way is end-time of the $P(k \mid k-1) = P(k-1 \mid k-1) + V(k)$ (15) last non-dropped attack request subtracting start-time of the first Correct (Measurement Update) non-dropped attack request during one attack interval. As we men- tion before, the end-to-end response time of a HTTP request in- $P(k \mid k-1)$ $K_д(k) =$ (16) volves three parts: the network latency, the queuing delay, and the $(P(k \mid k-1) + W(k))$

CPU NW $\hat{x}(k \mid k) = \hat{x}(k \mid k-1) + K_д(k)(z(k) - \hat{x}(k \mid k-1))$ (17) Setting V PD ρ(T) PM B (%) (MB/s) W/o W/o $P(k \mid k) = (1 - K_д(k))P(k \mid k-1)$ (18) (#) (ms) (%) (ms) Att. Att. att. att. where W(k) and V(k) are the covariances of w(k) and v(k), respec- EC2-111-2K 202 96.4 5.64 287 18.8 27.3 116 167 tively. In practice, we can estimate these two noise covariances us- ing automatic mathematical tools (e.g., autocovariance least-squares EC2-1212-4K 217 96.9 5.47 297 16.0 20.2 239 282 method [32]) or manual observation to tune the optimal value. $K_д(k)$ EC2-1414-6K 234 105.3 5.89 271 15.2 18.1 352 401 is termed the Kalman gain which represents the confidence index of Azure-111-1K the new measurement (z(k)) over the current estimate ($\hat{x}(k \mid k-1)$). 113 100.8 5.26 346 41.2 70.6 60 76 If $K_д(k)$ equals 1, it implies that the attacker totally trust the mea- Azure-1212-2K 137 99.1 5.56 479 34.6 58.3 119 141 surement, the effectiveness of adjusting the attack parameters in Azure-1414-3K 156 99.9 5.57 408 19.5 25.6 177 195 our attacks is totally depending on the accuracy of the estimated NSF-111-1K 97 101.0 5.40 453 49.1 76.8 58 72 value by Estimator. Using the Kalman filter, the Controller in Figure 5 can predict NSF-1212-2K 112 96.7 5.75 490 48.4 62.4 118 137 the required attack parameters at k-th burst given the historical NSF-1414-3K 131 99.4 5.22 470 34.6 51.0 173 186 results of all k-1 bursts, dynamically command the new parameters W/o att.: Without attacks, Att.: under Tail Attacks, to the bots, and automatically and effectively launch Tail Attacks. CPU: MySQL CPU usage, NW: Apache network traffic

Table 5: The corresponding parameters and bottleneck re- source utilization of Tail Attacks in real production settings. 5 REAL CLOUD PRODUCTION EVALUATION 5.1 Tail Attacks in Real Production Settings To evaluate the practicality of our feedback control attack frame- work in the real cloud production settings, we deploy a representa- prices and hardware configurations. However, the CPU core in EC2 tive benchmark website in the most popular two commercial cloud (2.40GHz Intel Xeon E5-2676 v3) is more powerful than the one platforms (Amazon EC2, Microsoft Azure) and one academic cloud in Azure (2.10GHz AMD Opteron 4171 HE or 2.40GHz Intel Xeon platform (Cloudlab[31]). E5-2673 v3). The worst one is in NSF Cloud (2.10GHz Intel Xeon Experiment Methodology. We adopt RUBBoS [34], a representa- E5-2450), where we run the VMs in Apt Cluster in the University tive n-tier web application benchmark modeled after the popular of Utah's Downtown Data Center . news website Slashdot. We configure RUBBoS using the typical 3- For the baseline workload, we have two chosen criteria: the tier or 4-tier architecture. A sample setting EC2-1414-6K in Table 5 bottleneck resource utilization (e.g., CPU utilization or Network is 1 Apache web server, 4 Tomcat application servers, 1 C-JDBC bandwidth) is less than %50 (Column 6 and 8 of Table 5), and no clustering load-balance server, 4 MySQL database servers deployed long-tail latency problem exists in without-attacks cases shown in in Amazon EC2, and 6000 concurrent legitimate users. RUBBoS has the Table 1. Because EC2 has more powerful CPU that we used in a workload generator to emulate the behavior of legitimate users our experiments, it can serve higher baseline workload than the to interact with the target benchmark website. Each user follows a other two. The network overhead can be the bottleneck resource Markov chain model to navigate among different webpages, with in EC2 platform even though CPU is at low utilizations [17]. In our averagely 7-second think time between every two consecutive re- experiments, MySQL CPU is the bottleneck tier in Azure and NSF quests. Through modifying the RUBBoS source code, we simulate Cloud due to the high resource consumption of database operations. various baseline workload (e.g., variable concurrent users during We pre-define our attack goal as the 95th percentile response the experiment). Meanwhile, in our experiments we adopt a central- time longer than 1 second and the utilization of the bottleneck ized strategy to coordinate and synchronize bots [12, 37, 49] (more resource less than 50%, and fix the attack interval as 2 seconds. Thus, discussion about distrubuted bots coordination and synchroniza- we need to control damage length P D longer than 100 milliseconds, tion in Section 7). All the bots are in the same location to rule out the millibottleneck length P M B less than 500 milliseconds. impact of the shift of network-latency. We control a small bot farm Results. Column 2 to 5 of Table 5 show the corresponding model of 10 machines (one of which serves as a centralized controller), parameters in our real cloud production setting experiments con- synchronized by NTP services, which can achieve millisecond pre- trolled by our attack framework. It clearly shows that our attacks cision [15]. Each bot uses Apache Bench to send intermittent bursts controlled by our algorithm can achieve the predefined targets (5% of attack HTTP requests, commanded by our control framework. drop ratio $\rho(T)$, damage length P D < 100ms, millibottleneck length All the VMs we run are 1 vCPU core and 2GB memory, which is P M B < 500ms). EC2 has more powerful CPU, so it requires more the basic computing unit for the commercial cloud providers. We attack requests per burst V to launch a successful attack. As the select HDD disk since our experimental workloads are browse-only servers scale out (from 111 to 1212, 1414), the n-tier system can CPU intensive transactions. We select t2.small instance ($0.023 service more legitimate users, at the same time, in order to launch per hour) in Amazon EC2 us-east-1a zone, and A1 ($0.024 per Tail Attacks it

requires higher V due to their higher capacity, which hour) instance in Microsoft Azure East US zone. They have similar means that bigger websites need larger botnet to attack. Column 7

400 0.25 1 300 0.2 mean=0.098 0.8 mean=0.22 Req. [#]

PMB [s] PD [s] 0.15 0.6 200 0.1 0.4 100 0.05 0.2 0 0 0 0 20 40 60 80 100 120 140 160 180 0 20 40 60 80 100 120 140 160 180 0 20 40 60 80 100 120 140 160 180 Timeline [s] Timeline [s] Timeline [s] (a) The requests of baseline workload vary (b) Damage length inferred by the prober (c) Millibottleneck length estimated by the from 2000 to 500 concurrent users. nearly equals the target 100ms. bots is less than 500ms, achieving the goal.

CPU Usage [%] Drop Ratio [%] 300 15 100 250 mean=5.5 80 mean=52 200 10 V [#]

60 150 100 5 40 50 20 0 0 0 0 20 40 60 80 100 120 140 160 180 0 20 40 60 80 100 120 140 160 180 0 20 40 60 80 100 120 140 160 180 Timeline [s] Timeline [s] Timeline [s] (d) Required attack volume V controlled (e) Drop ratio for legitimate users meets (f) Bottleneck resource utilization approxi- by our framework increases as back- the target 5%, indicating the effectiveness mately equals 50%, indicating the effective- ground requests decreases in Fig. 8a. controlled via damage length P D in Fig. 8b. ness controlled via P M B in Fig. 8c.

Figure 8: Results of Tail Attacks on RUBBoS web application in a real cloud production setting

and 9 of Table 5 show the average CPU utilization of MySQL tier and 5.2 Tail Attacks under IDS/IPS systems the average network traffic of Apache tier, which are the bottleneck Next, in order to evaluate the stealthiness of Tail Attacks under the resource in our experimental environment. Under our attacks in the radar of state-of-the-art IDS/IPS systems, we deploy some popu- real cloud production settings, the end users encounter the long-tail lar defense tools in the web tier in our RUBBoS environments to latency problem (see Table 1). However, all the bottleneck resources evaluate whether our attacks can be detected by them. are under moderate average resource utilization even in large scale Experiment Methodology. Typically, the popular solution to mit- case (e.g., CPU is 25.6% in Azure-1414-3K, and NW is 401MB/s in igate the application layer DDoS attacks is identifying abnormal EC2-1414-6K). These experimental results show that our attacks user-behavior [38, 44–47]. Snort [9] is a signature rules based Open- can cause significant long tail latency problem, while the servers Source IDS/IPS tool that is widely used in practice for DDoS de- are far from saturation, guaranteeing a high level stealthiness. fense, the users can customize the alert rules by setting reasonable Figure 8 further illustrates the 3-minute detailed experimental thresholds in the specific systems. We set alert rules following some results with various baseline workload under our attacks launched user-behavior models in Snort to evaluate whether our attacks devi- by our control framework in NSF-1212 setting. Figure 8a shows ate from the model (judging based on predefined thresholds). In the baseline workload changes from 2000 to 500 concurrent users at the following experiments, we configure 2000 concurrent legitimate middle point of the experiment (each user has 7-second think time users, and our attack goal is to achieve the 95th percentile response between two webpages). Note that the shift of baseline workload in time longer than 1s for these legitimate users. this case is different from the previous cases in which we scale out Results. To validate the user behavior model, we take request the servers. Figure 8d shows the required burst volume dynamically dynamics model [33] as an example. The authors analyzed the dis- adjusted by our framework, a sudden increase at the middle due to

tribution of a user's interaction with a Web server from a collection the decrease of baseline workload. Figure 8b and Figure 8c depict of Web server logs, categorized four session types (searching, brows- damage length and millibottleneck length estimated by the prober ing, relaxed and long session), and modeled the features for each and the bots, respectively. The measured average value of two criti- type of session based on average pause between sessions. Typically, cal metrics of our model are successfully controlled in the target average inter-request interval for searching and browsing sessions range. Figure 8e and Figure 8f depict the drop ratio during every is less than 10 seconds. RUBBoS [34] also models the inter-request attack interval and the CPU utilization of MySQL using 1 second interval per session as a Poisson process with the mean as 7, which granularity monitoring, respectively, which match our predefined means an average 7-second think time between two webpages for attack goal to a great extent. In general, through the RUBBoS ex- each session. In this case, we can calculate the 95% confidence in- periments in real cloud production settings, we can validate and terval as (2.814, 14.423). We can set the minimal boundary of the confirm the practicality of our attack control framework. interval (e.g., rounded to 3, the statistical granularity is seconds in

Alert Rule Parameters Triggered Alert Number Inter-request Model Botnet Designed Attack Pattern Threshold Sampling Period Bots Legitimate Users Actual Predicted minS minG 1G. 2G. 4G. 8G. 1 3s 0 19262 3 3 300 1 1/3s 2/6s 4/12s 8/24s 5 15s 0 7032 3 6 600 2 - 1/6s 2/12s 4/24s 10 30s 0 1074 3 12 1200 4 - - 1/12s 2/24s 15 45s 0 174 3 24 2400 8 - - - 1/24s 20 60s 0 23 1G.: 1Group, 2G.:2Groups, 4G.:4Groups, 8G.:8Groups 50 150s 0 2 minS.:minimal Size, minG.:minimal Group 100 300s 0 0 Table 7: The attacker conservatively predicts the inter- Table 6: The attacker can successfully predict the inter- request model. To achieve a burst with 300 requests in ev- request model. Less the sampling period, higher alerts from ery 3 seconds (V=300, L=50, I=3), more conservative predict the legitimate users, indicating higher false positive error. needs bigger botnet. However, the attacker can trigger no alert by carefully con- trolling the request pattern. thresholds of the user-behavior model, such that they can design a corresponding attack pattern to avoid be detected. Most user- behavior models are public to both the defenders and the attackers, Snort) as the alert threshold to validate whether our attacks deviate the only gap is the specific alert threshold and rules, which typically the RUBBoS model, since the smaller alert threshold can lead to less are learned from the server's past logs for their anomaly detection false positive error. We use the "detection_filter" property in Snort systems by the defenders of specific websites [33]. However, the at- to define the alert rules monitoring inter-request rate for each IP, tackers can use the questionnaire approach to similarly estimate the which generate an alert once the amount of requests from the same real users' behaviors, and use a conservative value as the potential IP exceeds the predefined threshold during a sampling statistical threshold to design the attack pattern with the price of increasing period. Column 1 and 2 of Table 6 depict the threshold and the more bots shown in Table 7. The four rows in Table 7 show four sampling interval for these alert rules, respectively. cases with different predicted values (3, 6, 12, 24). Obviously, as the We use RUBBoS client to simulate 2000 concurrent legitimate predicted value is more conservative, it requires a bigger botnet users and craftily control our bots catering to the inter-request (minimal size is 2400 when the prediction is 24). In the '24' case, the model in RUBBoS to bypass the above alert rules in Table 6 while attacker must split the 2400 bots into 8 groups, each group takes achieving our attack goal. Due to the limited numbers of IPs, we can turn to send a

burst of 300 requests in every 24-second interval. not have enough IP address to simulate the 2000 legitimate users and the bots from real IP address to evaluate the above detection rules in Snort. However, we can map a session to a individual IP address and implement the same detection algorithm using the 6 DETECTION AND DEFENSE “detection_filter" property of Snort to validate the above alert rules. Here, we consider a solution to detect and defend against Tail At- We conduct a 3-minute successful attack experiment in our RUBBoS tacks. There exists no easy approach to accurately distinguish the websites. In our case, to achieve the attack goal, the required attack attack requests from legitimate requests. Instead, we can identify parameters are V as 300, L as 50 and I as 3. To follow the inter- the attack requests by detecting the burst and matching the bursty request model in RUBBoS (alert threshold as 3), it requires 301 arrivals to millibottlenecks and cross tier queue overflown (Event2 totally-synchronized bots (one session simulates one bot in our and Event3 in Section 2). We present a workflow to mitigate our experiments and sends one request in more than 3s interval) while attacks involving three stages: fine-grained monitoring, burst de- avoid triggering the alerts (deviating from the model). Column tection, and bots blocking. 3 and 4 of Table 6 report the traced alert number from the bots Fine-grained monitoring. The unique feature of our attacks is and the legitimate users for these rules, respectively. As a result, that we exploit the new-found vulnerability of millibottlenecks our attacks can be totally invisible to these alert rules. Note that (with subsecond duration) in recent performance studies of web as the sampling interval increases the alerts from the legitimate applications deployed in cloud [41–43]. In order to capture milli- users decrease, the reasonable sampling interval can reduce the bottlenecks, the monitoring granularity should be less than the false positive errors (typically the interval should be on the order millibottlenecks period in millisecond level. For example, if the of minutes). Another important guide to our attacks is that as the monitoring granularity is 50ms, it can definitely pinpoint the milli- sampling interval increases, the threshold of the rules also has to bottleneck longer than 100ms, probably can seize the millibottle- increase, which can give us more flexible options to send the attack neck in the range of 50ms to 100ms, but absolutely can not capture requests using different intervals (e.g., in the above case, less than the millibottleneck less than 50ms. Thus, how to choose the mon- 10 requests every 30s interval per session, or less than 20 requests itoring granularity is depending on the observation and specific every 60s interval per session). To choose which sending pattern, requirement of eliminating the special millibottlenecks. we can further learn from the other user-behavior models to make Burst detection. Through fine-grained monitoring, we may ob- our attacks even more stealthy. serve a bunch of spike for each metrics (e.g., CPU utilization, request Someone may argue that the “no alert" results are got from traffic, queue usage, etc.). However, our purpose is to detect the the assumption that the attackers comprehensively know the alert burst of attack requests, so we must discriminate the actual attack

bursts from them. Based on the unique scenario of our attacks in the bar of an effective Tail attack since each web server only re- Section 2, we can define our attack bursts in which all the follow-ceives a portion of total attacking requests, thus higher volume of ing events occur simultaneously: very long response time requests attacking requests per burst are needed to create millibottlenecks (dropped requests), cross tier queue overflown, millibottlenecks in the system. On the other hand, if the bottleneck resource is in (e.g., CPU utilization, I/O waiting), and burst of requests. If all the a non-scalable tier such as the relational database tier, the load events are

observed in the same spike duration, we can regard the balancing in front tiers does not help mitigating the effectiveness spike duration as a potential attack burst. of Tail Attacks. This is because no matter which web server an IP-based statistical analysis defense.Once we identify the bursts attacking HTTP request arrives at, the database queries sent out by of Tail Attacks, the next task is to distinguish the requests of the the HTTP request may eventually converge to the same database bots from the requests of the legitimate users during the burst and server, and create millibottlenecks there. block them. The attacker, in our attack scenario, aims to coordinate Impact of cloud scaling. Large-scale web applications usually and synchronize the bots to sending bursts of attack requests during adopt dynamic scaling strategy (e.g., Amazon Auto Scaling [2]) for short “ON" burst period and repeat the process after long “OFF" better load balancing and resource efficiency, however, Tail Attacks burst period as introduced in Section 2, we can ideally introduce can easily bypass current dynamic scaling techniques, since the a new request metric that quantifies the suspicion index of the in- control window of the state-of-the-art scaling mechanism is usu- coming requests by aggregating the requests statistics during “ON" ally in minute-level (e.g., Amazon CloudWatch monitoring in a burst and “OFF" burst for further analysis. Specially, we define the minute granularity), while Tail Attacks are too short (sub-second suspicion index for each IP address as follows: duration) for them to catch and take any scaling actions [40]. The NBI P main advantage of Tail attacks is that it is invisible to most mon- SII P = (19) itoring programs and can remain hidden for a long time because NI P of the low volume characteristic and the sub-second duration of where NBI P and NI P are the number of requests for each IP during millibottlenecks as shown in our experimental results. “ON" burst and the attack interval T (including “ON" and “OFF" Browser compatibility check. Some state-of-the-art defense burst), respectively. If SII P for a IP is close to 1, the IP is likely tools may check the header information of each HTTP request to be a bot; on the other hand, SII P of the legitimate user can be to determine whether it is sent from a real browser or from a bot. approximately the target attack drop ratio (e.g., 0.5 if the target is A HTTP request sent from a real browser usually has completed 95th percentile response time longer than 1 second). Figure 9 shows header information such as “User-Agent”, while such information Probability Density Function of SII P in the RUBBoS experiment may not appear or be difficult to be generated by a bot which only in Section 5.1. The red bar represents the bots and the green bars uses a script language to generate HTTP requests. In addition, some represent the 2000 legitimate users. In this way to identify bots, the websites such as Facebook need a legitimate user to login first be- false positive and false negative error can be close to 0 with 100% fore any following transactions, especially heavy query requests high precision. (detailed explanations in Section 4.2). They may track the cookies stored in the client side browser in order to keep an active session in Number of IPs (Sessions)

350 the server side; a bot may not be able to interact with such websites 300 Normal Users due to the lack of support of a real browser. We can address these 250 Attackers challenges by using PhantomJS [36] to generate attacking HTTP 200 requests. PhantomJS is a headless web browser without a heavy 150 graphical user interface. It is built on top of WebKit, the engine 100 behind Chrome and Safari. So PhantomJS can behave the same as a 50 0 real browser does. Therefore, an attacker can launch browser-based 0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1 Tail Attacks using heavy requests as attack requests by PhantomJS, SIIP and the generated requests will be extremely difficult to distinguish from the requests sent by legitimate users. Figure 9: Identify bots Distributed bots coordination and synchronization. One pre- condition of Tail attacks is that bots

could be coordinated and synchronized so that the generated HTTP requests are able to reach the target web system within a short time window. Many 7 DISCUSSIONS previous research efforts already provide solutions, using either Impact of load balancing. Some web applications adopt load centralized [12, 37, 49] or decentralized methods [22], to coordi- balancing (e.g., Amazon Elastic Load Balancing [4]) in front of the nate bots to send synchronized traffic to cause network congestion system to distribute load across multiple web servers. This type of at a specific network link. Centralized control can achieve higher load balancing works well for stateless servers such as web servers, level of bots coordination and synchronization, which enables a in which incoming traffic are supposed to evenly distribute among more effective Tail Attack compared to decentralized methods. In them. However, whether or not load balancing can mitigate the ef- this paper we adopt the centralized control method to do experi- fectiveness of Tail attacks depends on the location of the bottleneck ments. On the other hand, a decentralized method in general is able resource in the target web system. For example, if the bottleneck to coordinate and synchronize more bots than a centralized one, resource is in the web server tier, load balancing indeed increases thus making it possible to target large-scale/high-capacity websites.

However, a decentralized method is more challenging to control with a higher level of stealthiness. To thoroughly comprehend the the length of each burst of attacking requests arrived at the target attack scenario (Section 2), we formulated the impact of our attacks website, thus mitigating the effectiveness of Tail Attacks. in n-tier systems based-on queueing network model, which can effectively guide our attacks in a stealthy way (Section 3). We imple- mented a feedback control-theoretic (e.g., Kalman filter) framework that allows attackers to fit the dynamics of background requests 8 RELATED WORK or system state by dynamically adjusting the optimal attack pa- In this section, we review the most relevant work with regard to low- rameters (Section 4). To validate the practicality of our attacks, we volume application-layer attacks, which is even stealthier to avoid evaluated our attacks through not only analytical, numerical, and traditional network-layer based detection mechanisms [30, 35, 48]. simulation results but also benchmark web applications equipped Low-volume Application Layer DDoS attacks. Low-volume with state-of-the-art DDoS defense tools in real cloud production DDoS application attacks are characterized by a small number settings (Section 3 and Section 5). We further proposed a solution of attack requests transmitted strategically to the target appli- to detect and defense the proposed attacks, involving three stages: cation servers, as an extension of network-layer low-volume at- fine-grained monitoring, identifying bursts, and blocking bots (Sec- tacks [12, 14, 21, 22, 25, 27, 39]. Macia-Fernandez et al. initially pro- tion 6). More generally, our work is an important contribution posed low-rate attacks against application servers (LoRDAS) [28] towards a comprehensive understanding of emerging low-volume that send traffic in periodic short-time pulses at a low rate, shar- application DDoS attacks. ing the similar on-off attack pattern with our attacks. Slow-rate attacks [8] deplete system resources on the server's side by sending A DERIVATION OF MILLIBOTTLENECK (e.g., slow send/Slowloris [13]) or receiving (e.g., slow read) traffic LENGTH at a slow rate. Our attacks share the similar features of low attack During millibottleneck length, the bottleneck resources sustain sat- volume with low-rate and slow-rate attacks. uration. Millibottleneck length should involve the serving time for However, compared to these two attacks, our attacks dig more both attack and normal requests during a burst. The attackers only deeply into n-tier architecture applications while LoRDAS attacks send requests within the short "ON"

period, thus the amount of at- only in 1-tier application server. Our analytical model for n-tier sys- tack requests is the burst volume V. Meanwhile, the legitimate users tems can guide and guarantee our attacks dynamically controlled in always send requests during millibottleneck length, thus the satu- a more effective and elusive way by accurately estimating damage ration length is an infinite recursive process until it converges to length and millibottleneck length. In addition, slow-rate attacks need zero exponentially. Equation (20) represents millibottleneck length to develop well-crafted HTTP header (Slow Headers) or body (Slow derived through the geometric progression in mathematics, here, Body) thus expose obvious attack patterns to the defense tools, m limits to infinity. while Tail Attacks use the legitimate and normal heavy requests as our attack requests thus hide deeper. More importantly, we ex- 1 1 1 ploit the ubiquity of millibottlenecks (with sub-second duration) PMB = V +V λn C n, A C n, A C n, L and strong dependencies among distributed nodes for web appli- 1 1 1 cations, leading to long-tail latency problem with a higher level of +V λn λn + ... stealthiness than LoRDAS and Slow-rate attacks. C n, A C n, L C n, L Detecting and Defending against Application DDoS attacks. 1 1 +V (λn)m To mitigate application DDoS attacks, existing solutions typically C n, A C n, L focus on distinguishing application-layer DDoS traffic from the m X 1 1 = lim V (λn)k (20) traffic created by legitimate users, such as abnormal user-behavior m ∞ C n, A C n, L in high-level features [44, 46, 47] of surging web pages, in session- k =0 level [38], or in request-level [45]. To be stealthy, for the features of (1 − (λn C) 1) m+1 1 =V n, L navigating web pages, our attacks can learn from the user behaviors lim C n, A m ∞ (1 − (λn 1)) of a legitimate user; as for session or requests level of our attacks, we C n, L can calculate the required optimized attack volume and botnet size 1 1 =V C n, A (1 − (λn 1 C n, L)) discuss in Section 5. Compared to existing solutions, our proposed countermeasure can be tied to the unique feature of our attacks and accurately capture the bots. where 1/Cn,A and 1/Cn, L are the service time for attack and normal requests in the bottleneck tier, respectively.

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Chair endowment. Any opinions, findings, and conclusions or rec- computer communications (SIGCOMM'03). ACM, Karlsruhe, Germany, 75–86. ommendations expressed in this material are those of the author(s) [26] Chien-An Lai, Josh Kimball, Tao Zhu, Qingyang Wang, and Calton Pu. 2017. milliScope: a Fine-Grained Monitoring Framework for Performance Debugging and do not necessarily reflect the views of the National Science of n-Tier Web Services. In Proceedings of the IEEE 37th International Conference Foundation or other funding agencies and companies mentioned on Distributed Computing Systems (ICDCS'17). IEEE, Atlanta, GA, USA, 92–102. [27] Xiapu Luo and Rocky KC Chang. 2005. On a New Class of Pulsing Denial-of- above. Service Attacks and the Defense. In Proceedings of Network and Distributed System Security Symposium

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